higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T160**(E)**(A12)T
NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N4

(8060034)

12 April 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 5 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL SURVEYING N4
TIME: 3 HOURS
MARKS:100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Define each of the following:

1.1 Vertical angles

1.2 Catenary

1.3 Slope

1.4 Standard temperature

1.5 Coefficient of expansion

(5 × 2) **[10]**

QUESTION 2

2.1 What is the purpose of surveying? (4)

2.2 State FIVE purposes of fieldwork in surveying. (5)

2.3 Name FIVE requirements for a magnetic needle. (5)

[14]

QUESTION 3

3.1 Draw the South African co-ordinates system. Clearly show in which quadrants the x-and y-co-ordinates are positive and in which quadrants they are negative. (6)

3.2 The length of a rectangular plot is twice the breadth, and the area covers 6 hectares.

Calculate the length and breadth of the plot and then make a drawing to scale 1 mm:10 m. Show ALL the dimensions (6)

3.3 Calculate the area of a plot with an irregular shape with the ordinates 8, 7, 9, 6, 5, 3 and 0, all in m. The width between the ordinates is 10 m. (4)

[16]

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QUESTION 4

4.1 What do GN and RB stand for? (2)

4.2 Differentiate between each of the following:

4.2.1 Quadrant bearing and whole-circle bearing

4.2.2 True bearing and magnetic bearing

4.2.3 True north and magnetic north

(3 × 2) (6)

4.3 Convert the following whole-circle bearings (WCB) to quadrant bearings (QB):

4.3.1 25° 40' (2)

4.3.2 225° 26' (3)

4.3.3 140° 50' (3)

4.3.4 340° 50' (3)

4.4 The magnetic bearing of line AB is 143° 40'.

Determine the true bearing if the declination is 8° 19' W. (3)

4.5 Convert a QB of S 38248 45' E. (2)

[24]

QUESTION 5

5.1 Describe the hand signals a surveyor uses to convey the following instructions:

5.1.1 Move well to the right

5.1.2 Hurry

5.1.3 Keep the staff steady in position

5.1.4 Continue to move to the left

(4 × 2) (8)

5.2 Explain step-chaining by using a diagram. (6)

[14]

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QUESTION 6

6.1 A distance is measured at a slope of -7°15' and found to be 196,54 m at a temperature of 11,5 °C.

Calculate the correct horizontal distance if the tape was standardised at 20 °C and e = 0,0000112/°C. (5)

6.2 A length of 95,45 m was measured between two points at an angle of 7° 40' 20" from the horizontal.

Calculate the horizontal distance between the points. (5)

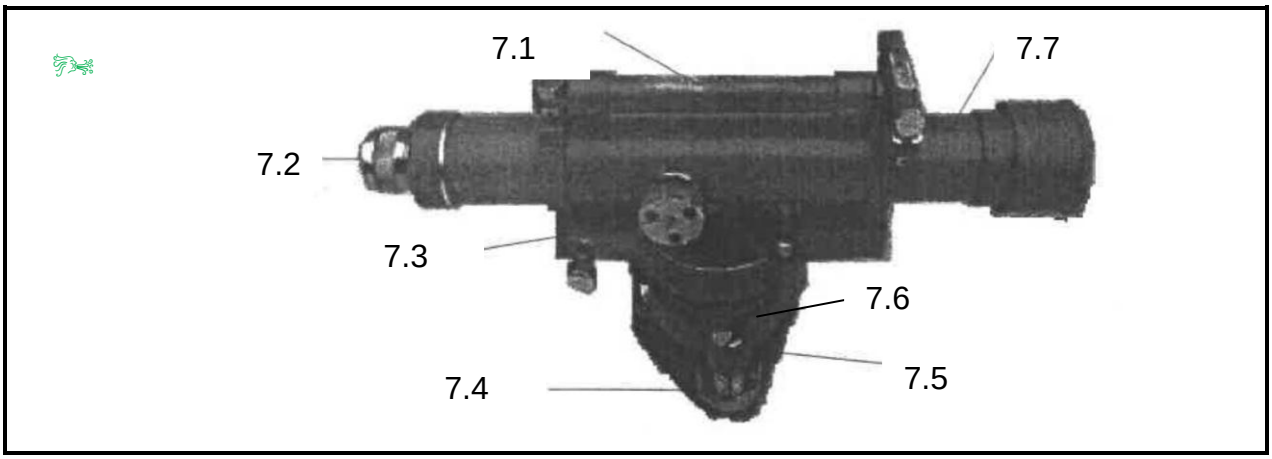
6.3 A length of 272 m was measured with a steel tape at 2 300 m above sea level. The radius of the earth is 6 373 km.

Calculate the correct distance. (5)

[15]

QUESTION 7

Label the numbered parts of the dumpy level below by writing write only the name of the part next to the question number (7.1–7.7) in the ANSWER BOOK.



DUMPY LEVEL

(7 × 1) **[7]**

TOTAL: 100

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BUILDING AND STRUCTURAL SURVEYING N4

FORMULA SHEET

Any other applicable formula may also be used.

1. $A = {}^{1}/_{2}\, bx \perp h$

2. $A = \sqrt{s(s-a)(s-b)(s-c)}\,,\ \frac{a+b=c}{2}$

3. $A = \frac{(a+b)x\ h}{2}$

4. $A = a \text{ x } h$

5. $A = \frac{d}{3}[(y_1 + y_n) + 2(y_3 + y_5 + \ldots + (y_n - 2) + 4(y_2 + y_4 + \ldots + y_n - 1)]$

6. $A = d\ [(\frac{y_2 + y_n}{2} + y_2 + y_3 + \ldots + y_{n-1}]$

7. $C = L \text{ x } e \text{ x} (t_m - t_s)$

8. $C = \frac{W^2 L^3}{24T^2}$

9. $C = L\ (1 - \text{Cos}\theta)$

10. $C = H\ (\text{sec} - 1)$11. $C = \frac{LH}{R}$

12. $A = \frac{1}{2} abSinC$

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